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**Database Practical
assignment**

**Rosebank International
college**

STEP 1 — CREATE THE DATABASE AND TABLES (10 marks)

```
CREATE DATABASE ArtGalleryDB_AssignmentB;
```

```
GO
```

```
USE ArtGalleryDB_AssignmentB;
```

```
GO
```

```
CREATE TABLE Artist (
```

```
    ArtistID INT PRIMARY KEY,
```

```
Name VARCHAR(50) NOT NULL,  
Surname VARCHAR(50) NOT NULL  
);  
GO
```

```
CREATE TABLE Genre (  
    GenreID INT PRIMARY KEY,  
    Description VARCHAR(150) NOT NULL  
);  
GO
```

```
CREATE TABLE Artwork (  
    ArtworkID INT PRIMARY KEY,  
    GenreID INT NOT NULL,  
    ArtistID INT NOT NULL,  
    Title VARCHAR(150) NOT NULL,  
    CONSTRAINT FK_Artwork_Genre FOREIGN KEY (GenreID) REFERENCES Genre(GenreID),  
    CONSTRAINT FK_Artwork_Artist FOREIGN KEY (ArtistID) REFERENCES Artist(ArtistID)  
);  
GO
```

```
CREATE TABLE Exhibition (  
    ExhibitionID INT PRIMARY KEY,  
    Description VARCHAR(150) NOT NULL  
);  
GO
```

```
CREATE TABLE Entry (  
    EntryID INT PRIMARY KEY,  
    ArtworkID INT NOT NULL,  
    ExhibitionID INT NOT NULL,
```

```
CONSTRAINT FK_Entry_Artwork FOREIGN KEY (ArtworkID) REFERENCES Artwork(ArtworkID),  
CONSTRAINT FK_Entry_Exhibition FOREIGN KEY (ExhibitionID) REFERENCES  
Exhibition(ExhibitionID),  
CONSTRAINT UQ_Entry_Artwork_Exhibition UNIQUE (ArtworkID, ExhibitionID)  
);  
GO
```

STEP 2 — INSERT RELEVANT DATA (10 marks)

Record-count check: Artist = 5 (minimum 5); Genre = 3 (minimum 3); Artwork = 20 (minimum 20); Exhibition = 15 (minimum 15); Entry = 30 (and several artworks appear in multiple exhibitions).

```
INSERT INTO Artist (ArtistID, Name, Surname) VALUES
```

```
(1, 'Aisha', 'Mthembu'),  
(2, 'Ethan', 'Botha'),  
(3, 'Lerato', 'Dlamini'),  
(4, 'Sipho', 'Van Wyk'),  
(5, 'Mia', 'Daniels');
```

GO

INSERT INTO Genre (GenreID, Description) VALUES

(1, 'Abstract'),
(2, 'Landscape'),
(3, 'Still Life');

GO

INSERT INTO Artwork (ArtworkID, GenreID, ArtistID, Title) VALUES

(1, 3, 2, 'After the Storm'),
(2, 1, 4, 'Autumn Table'),
(3, 2, 1, 'City Pulse'),
(4, 3, 3, 'Coastal Dawn'),
(5, 1, 5, 'Crimson Shapes'),
(6, 2, 2, 'Distant Hills'),
(7, 3, 4, 'Garden Study'),
(8, 1, 1, 'Golden Harvest'),
(9, 2, 3, 'Harbour Morning'),
(10, 3, 5, 'Light and Shadow'),
(11, 1, 2, 'Midnight Street'),
(12, 2, 5, 'Morning Market Deluxe'),
(13, 3, 1, 'Old Oak'),
(14, 1, 3, 'Quiet Harbour'),
(15, 2, 5, 'Red Earth'),
(16, 3, 2, 'Summer Fields'),
(17, 1, 4, 'The Blue Vase'),
(18, 2, 1, 'Urban Stories'),
(19, 3, 3, 'Watercolour Rain'),
(20, 1, 5, 'Yellow Horizon');

GO

INSERT INTO Exhibition (ExhibitionID, Description) VALUES

(1, 'Art in Motion'),
(2, 'Autumn Arts Week'),

(3, 'Coastal Collection'),
(4, 'Creative Futures'),
(5, 'Everyday Objects'),
(6, 'Foundations'),
(7, 'Gallery Open Day'),
(8, 'Great South African Art'),
(9, 'Light and Colour'),
(10, 'Local Voices'),
(11, 'Masters and Mentors'),
(12, 'Nature and Place'),
(13, 'Spring Showcase'),
(14, 'The Collector''s Room'),
(15, 'Young Artists');

GO

INSERT INTO Entry (EntryID, ArtworkID, ExhibitionID) VALUES

(1, 1, 1),
(2, 2, 3),
(3, 3, 5),
(4, 3, 10),
(5, 4, 7),
(6, 4, 12),
(7, 5, 9),
(8, 6, 11),
(9, 7, 13),
(10, 7, 3),
(11, 8, 15),
(12, 9, 2),
(13, 9, 7),
(14, 10, 4),
(15, 11, 6),
(16, 11, 11),

(17, 12, 8),
(18, 13, 10),
(19, 14, 12),
(20, 15, 14),
(21, 15, 4),
(22, 16, 1),
(23, 17, 3),
(24, 18, 5),
(25, 18, 10),
(26, 19, 7),
(27, 20, 9),
(28, 20, 14);
GO

STEP 3 — UPDATE STATEMENT (10 marks)

The artwork title is updated by ArtworkID, demonstrating a precise one-record UPDATE.

UPDATE Artwork

SET Title = 'Morning Market Deluxe'

WHERE ArtworkID = 12;

STEP 4 — DELETE STATEMENT (10 marks)

An Entry record is deleted by EntryID, demonstrating a controlled DELETE while preserving the related Artwork and Exhibition.

```
DELETE FROM Entry
```

```
WHERE EntryID = 30;
```

STEP 5 — ARTWORK, ARTIST AND GENRE REPORT (15 marks)

Purpose of report: To provide a complete artwork catalogue showing every artwork title, its artist and its genre. The report is sorted alphabetically by genre first, followed by artwork title.

SELECT

g.Description AS Genre,

a.Title AS ArtworkTitle,

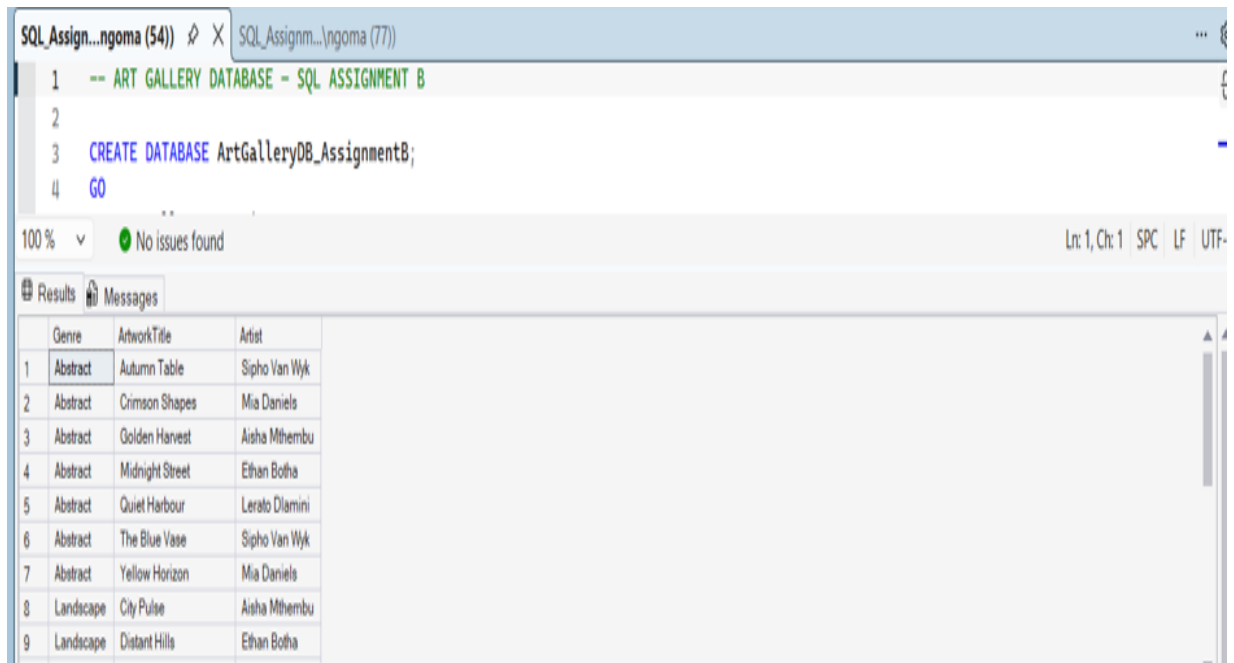
CONCAT(ar.Name, ' ', ar.Surname) AS Artist

FROM Artwork AS a

INNER JOIN Artist AS ar ON a.ArtistID = ar.ArtistID

INNER JOIN Genre AS g ON a.GenreID = g.GenreID

ORDER BY g.Description ASC, a.Title ASC;



The screenshot shows a SQL IDE interface. The top pane displays a SQL script for creating a database and inserting data. The bottom pane shows the results of a query in a table format.

SQL Script:

```
-- ART GALLERY DATABASE - SQL ASSIGNMENT B
CREATE DATABASE ArtGalleryDB_AssignmentB;
GO
```

Results Table:

	Genre	ArtworkTitle	Artist
1	Abstract	Autumn Table	Sipho Van Wyk
2	Abstract	Crimson Shapes	Mia Daniels
3	Abstract	Golden Harvest	Aisha Mthembu
4	Abstract	Midnight Street	Ethan Botha
5	Abstract	Quiet Harbour	Lerato Dlamini
6	Abstract	The Blue Vase	Sipho Van Wyk
7	Abstract	Yellow Horizon	Mia Daniels
8	Landscape	City Pulse	Aisha Mthembu
9	Landscape	Distant Hills	Ethan Botha

STEP 6 — GROUP BY REPORT (15 marks)

Purpose of report: To summarise how many artworks belong to each genre. This helps the gallery understand the size of each genre category and demonstrates grouping and aggregate counting.

SELECT

g.Description AS Genre,

COUNT(a.ArtworkID) AS NumberOfArtworks

FROM Genre AS g

LEFT JOIN Artwork AS a

ON g.GenreID = a.GenreID

GROUP BY g.Description

ORDER BY g.Description ASC;

	Genre	NumberOfArtworks
1	Abstract	7
2	Landscape	6
3	Still Life	7

STEP 7 — HAVING REPORT (15 marks)

Purpose of report: To identify only the genres containing at least six artworks. HAVING is appropriate because the condition is applied to the grouped COUNT result rather than to individual artwork rows.

```
SELECT
    g.Description AS Genre,
    COUNT(a.ArtworkID) AS NumberOfArtworks
FROM Genre AS g
INNER JOIN Artwork AS a
    ON g.GenreID = a.GenreID
GROUP BY g.Description
HAVING COUNT(a.ArtworkID) >= 6
ORDER BY NumberOfArtworks DESC, g.Description ASC;
```

	Genre	NumberOfArtworks
1	Abstract	7
2	Landscape	6
3	Still Life	7

STEP 8 — JOIN REPORT (15 marks)

Purpose of report: To show which artworks are entered in which exhibitions, together with each artwork's artist and genre. This demonstrates joins across the Entry, Exhibition, Artwork, Artist and Genre tables.

SELECT

e.Description AS Exhibition,

a.Title AS ArtworkTitle,

CONCAT(ar.Name, ' ', ar.Surname) AS Artist,

g.Description AS Genre

FROM Exhibition AS e

INNER JOIN Entry AS en ON e.ExhibitionID = en.ExhibitionID

INNER JOIN Artwork AS a ON en.ArtworkID = a.ArtworkID

INNER JOIN Artist AS ar ON a.ArtistID = ar.ArtistID

INNER JOIN Genre AS g ON a.GenreID = g.GenreID

ORDER BY e.Description ASC, a.Title ASC;

	Exhibition	ArtworkTitle	Artist	Genre
1	Art in Motion	After the Storm	Ethan Botha	Still Life
2	Art in Motion	Summer Fields	Ethan Botha	Still Life
3	Autumn Art...	Harbour Mor...	Lerato Dila...	Lands...
4	Coastal Co...	Autumn Table	Sipho Van ...	Abstract
5	Coastal Co...	Garden Study	Sipho Van ...	Still Life
6	Coastal Co...	The Blue Vase	Sipho Van ...	Abstract
7	Creative Fu...	Light and Sha...	Mia Daniels	Still Life
8	Creative Fu...	Red Earth	Mia Daniels	Lands...
9	Everyday O...	City Pulse	Aisha Mith...	Lands...
10	Everyday O...	Urban Stories	Aisha Mith...	Lands...
11	Foundations	Midnight Street	Ethan Botha	Abstract
12	Gallery On...	Coastal Dawn	Lerato Dila...	Still Life

Query executed successfully. | LAPTOP-P9U04UCP\SQL EXPRESS ... | LAPTOP-P9U04UCP\ngoma ... | ArtGalleryDB_AssignmentB | 00:00:07 | Row: 1, Col: 1 | 54 rows

Ready

Reference list

Microsoft, 2026d. *SELECT - GROUP BY clause (Transact-SQL)*. Microsoft Learn. Available at: [Microsoft Learn – GROUP BY clause](#) (Accessed: 17 August 2026).

Microsoft, 2026e. *SELECT - HAVING clause (Transact-SQL)*. Microsoft Learn. Available at: [Microsoft Learn – HAVING clause](#) (Accessed: 17 August 2026).